

ARM-based Microcontroller

LPC2119/2129/2194/2292/2294

REGISTER DESCRIPTION

Each Timer contains the registers shown in Table 157. More detailed descriptions follow.

Table 157: TIMER0 and TIMER1 Register Map

Generic Name	Description	Access	Reset Value*	TIMER0 Address & Name	TIMER1 Address & Name
IR	Interrupt Register. The IR can be written to clear interrupts. The IR can be read to identify which of eight possible interrupt sources are pending.	R/W	0	0xE0004000 T0IR	0xE0008000 T1IR
TCR	Timer Control Register. The TCR is used to control the Timer Counter functions. The Timer Counter can be disabled or reset through the TCR.	R/W	0	0xE0004004 T0TCR	0xE0008004 T1TCR
TC	Timer Counter. The 32-bit TC is incremented every PR+1 cycles of pclk. The TC is controlled through the TCR.	RW	0	0xE0004008 T0TC	0xE0008008 T1TC
PR	Prescale Register. The TC is incremented every PR+1 cycles of pclk.	R/W	0	0xE000400C T0PR	0xE000800C T1PR
PC	Prescale Counter. The 32-bit PC is a counter which is incremented to the value stored in PR. When the value in PR is reached, the TC is incremented.	R/W	0	0xE0004010 T0PC	0xE0008010 T1PC
MCR	Match Control Register. The MCR is used to control if an interrupt is generated and if the TC is reset when a Match occurs.	R/W	0	0xE0004014 T0MCR	0xE0008014 T1MCR
MR0	Match Register 0. MR0 can be enabled through the MCR to reset the TC, stop both the TC and PC, and/or generate an interrupt every time MR0 matches the TC.	R/W	0	0xE0004018 T0MR0	0xE0008018 T1MR0
MR1	Match Register 1. See MR0 description.	R/W	0	0xE000401C T0MR1	0xE000801C T1MR1
MR2	Match Register 2. See MR0 description.	R/W	0	0xE0004020 T0MR2	0xE0008020 T1MR2
MR3	Match Register 3. See MR0 description.	R/W	0	0xE0004024 T0MR3	0xE0008024 T1MR3
CCR	Capture Control Register. The CCR controls which edges of the capture inputs are used to load the Capture Registers and whether or not an interrupt is generated when a capture takes place.	R/W	0	0xE0004028 T0CCR	0xE0008028 T1CCR
CR0	Capture Register 0. CR0 is loaded with the value of TC when there is an event on the CAP0.0(CAP1.0) input.	RO	0	0xE000402C T0CR0	0xE000802C T1CR0
CR1	Capture Register 1. See CR0 description.	RO	0	0xE0004030 T0CR1	0xE0008030 T1CR1
CR2	Capture Register 2. See CR0 description.	RO	0	0xE0004034 T0CR2	0xE0008034 T1CR2
CR3	Capture Register 3. See CR0 description.	RO	0	0xE0004038 T0CR3	0xE0008038 T1CR3
EMR	External Match Register. The EMR controls the external match pins MAT0.0-3 (MAT1.0-3).	R/W	0	0xE000403C T0EMR	0xE000803C T1EMR

*Reset Value refers to the data stored in used bits only. It does not include reserved bits content.

Interrupt Register (IR: TIMER0 - T0IR: 0xE0004000; TIMER1 - T1IR: 0xE0008000)

The Interrupt Register consists of four bits for the match interrupts and four bits for the capture interrupts. If an interrupt is generated then the corresponding bit in the IR will be high. Otherwise, the bit will be low. Writing a logic one to the corresponding IR bit will reset the interrupt. Writing a zero has no effect.

Table 158: Interrupt Register (IR: TIMER0 - T0IR: 0xE0004000; TIMER1 - T1IR: 0xE0008000)

IR	Function	Description	Reset Value
0	MR0 Interrupt	Interrupt flag for match channel 0.	0
1	MR1 Interrupt	Interrupt flag for match channel 1.	0
2	MR2 Interrupt	Interrupt flag for match channel 2.	0
3	MR3 Interrupt	Interrupt flag for match channel 3.	0
4	CR0 Interrupt	Interrupt flag for capture channel 0 event.	0
5	CR1 Interrupt	Interrupt flag for capture channel 1 event.	0
6	CR2 Interrupt	Interrupt flag for capture channel 2 event.	0
7	CR3 Interrupt	Interrupt flag for capture channel 3 event.	0

Timer Control Register (TCR: TIMER0 - T0TCR: 0xE0004004; TIMER1 - T1TCR: 0xE0008004)

The Timer Control Register (TCR) is used to control the operation of the Timer Counter.

Table 159: Timer Control Register (TCR: TIMER0 - T0TCR: 0xE0004004; TIMER1 - T1TCR: 0xE0008004)

TCR	Function	Description	Reset Value
0	Counter Enable	When one, the Timer Counter and Prescale Counter are enabled for counting. When zero, the counters are disabled.	0
1	Counter Reset	When one, the Timer Counter and the Prescale Counter are synchronously reset on the next positive edge of pclk. The counters remain reset until TCR[1] is returned to zero.	0

Timer Counter (TC: TIMER0 - T0TC: 0xE0004008; TIMER1 - T1TC: 0xE0008008)

The 32-bit Timer Counter is incremented when the Prescale Counter reaches its terminal count. Unless it is reset before reaching its upper limit, the TC will count up through the value 0xFFFFFFFF and then wrap back to the value 0x00000000. This event does not cause an interrupt, but a Match register can be used to detect an overflow if needed.

Prescale Register (PR: TIMER0 - T0PR: 0xE000400C; TIMER1 - T1PR: 0xE000800C)

The 32-bit Prescale Register specifies the maximum value for the Prescale Counter.

Prescale Counter Register (PC: TIMER0 - T0PC: 0xE0004010; TIMER1 - T1PC: 0xE0008010)

The 32-bit Prescale Counter controls division of pclk by some constant value before it is applied to the Timer Counter. This allows control of the relationship of the resolution of the timer versus the maximum time before the timer overflows. The Prescale Counter is incremented on every pclk. When it reaches the value stored in the Prescale Register, the Timer Counter is incremented and the Prescale Counter is reset on the next pclk. This causes the TC to increment on every pclk when PR = 0, every 2 pclks when PR = 1, etc.

Match Registers (MR0 - MR3)

The Match register values are continuously compared to the Timer Counter value. When the two values are equal, actions can be triggered automatically. The action possibilities are to generate an interrupt, reset the Timer Counter, or stop the timer. Actions are controlled by the settings in the MCR register.

Match Control Register (MCR: TIMER0 - T0MCR: 0xE0004014; TIMER1 - T1MCR: 0xE0008014)

The Match Control Register is used to control what operations are performed when one of the Match Registers matches the Timer Counter. The function of each of the bits is shown in Table 160.

Table 160: Match Control Register (MCR: TIMER0 - T0MCR: 0xE0004014; TIMER1 - T1MCR: 0xE0008014)

MCR	Function	Description	Reset Value
0	Interrupt on MR0	When one, an interrupt is generated when MR0 matches the value in the TC. When zero this interrupt is disabled.	0
1	Reset on MR0	When one, the TC will be reset if MR0 matches it. When zero this feature is disabled.	0
2	Stop on MR0	When one, the TC and PC will be stopped and TCR[0] will be set to 0 if MR0 matches the TC. When zero this feature is disabled.	0
3	Interrupt on MR1	When one, an interrupt is generated when MR1 matches the value in the TC. When zero this interrupt is disabled.	0
4	Reset on MR1	When one, the TC will be reset if MR1 matches it. When zero this feature is disabled.	0
5	Stop on MR1	When one, the TC and PC will be stopped and TCR[0] will be set to 0 if MR1 matches the TC. When zero this feature is disabled.	0
6	Interrupt on MR2	When one, an interrupt is generated when MR2 matches the value in the TC. When zero this interrupt is disabled.	0
7	Reset on MR2	When one, the TC will be reset if MR2 matches it. When zero this feature is disabled.	0
8	Stop on MR2	When one, the TC and PC will be stopped and TCR[0] will be set to 0 if MR2 matches the TC. When zero this feature is disabled.	0
9	Interrupt on MR3	When one, an interrupt is generated when MR3 matches the value in the TC. When zero this interrupt is disabled.	0
10	Reset on MR3	When one, the TC will be reset if MR3 matches it. When zero this feature is disabled.	0
11	Stop on MR3	When one, the TC and PC will be stopped and TCR[0] will be set to 0 if MR3 matches the TC. When zero this feature is disabled.	0